Vibe Theory

The Technical Companion: The Hyperbolic Substrate and the Findings

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June 7, 2026

Abstract

This is the technical companion to the conceptual account of Vibe Theory. Where the conceptual paper tells the story in plain terms, this one walks through the machinery and the findings: the finite, seeded testbed, the geometry of the substrate, and the eighty-three known-answer experiments that build the model from the bottom up. The substrate is the centerpiece. It is a growing hyperbolic crystal, a regular honeycomb of a Coxeter reflection group, and we treat it in detail: the Coxeter construction and the regular tessellations $\{p, q\}$ of the hyperbolic plane, the heptagrid $\{7, 3\}$, the regular honeycombs of hyperbolic space and the dodecagrid $\{5, 3, 4\}$ that is the model's real three-dimensional substrate, the classification that forces the spatial dimension into the window two-to-four and then to three, the Lorentz safety that comes from curvature rather than disorder, and the deterministic automaton that grows the crystal one cell at a time, forever. On that substrate sit the ternary tones and one local rule, from which we recover emergent geometry and time, the field and gauge content, a conserving Einstein equation and a discrete graviton, the quantum pillars with a derived Born rule, a cosmology with a dark sector, and sharp predictions that pass current bounds while excluding a lattice. We add the Standard Model results, anomaly-forced charge quantization, the Sakharov conditions, and the mass hierarchy from hyperbolic overlaps, and we mark honestly what is open. This is a simulable model with a concrete build target, not a finished empirical theory.

Status and caveats. This is exploratory work, a feasibility study rather than a validated result. The aim is to see whether a model of this kind can be made to hang together at all, and so far it looks promising. Much validation and further work remain. Several of the experiments reported here are preliminary, and some may turn out to be inaccurate or improperly set up. Nothing here should be read as established or taken too rigorously at this stage. It is offered in the spirit of curiosity and invitation, a direction that looks worth exploring rather than a finished theory or a claim of discovery.

1. Introduction

This paper is the technical companion to the conceptual account of Vibe Theory (Pollard, 2026a). The conceptual paper states what the theory says and how to picture it, deliberately light on mathematics. This one carries the machinery: the method, the geometry, and a walk through the full set of findings, eighty-three in all, from a finite and reproducible testbed (Pollard, 2026b). For each finding we say what was measured, what came out, and where it stands.

The thesis, stated once, is that reality is one experiential substance, the vibe, and that every other thing is a pattern of vibes and their relations on a growing, hyperbolic, causal mesh, updated by one local rule. The conceptual paper defends that picture. Here we test it. The method is to turn each load-bearing question into a runnable measurement, a finite seeded program that returns a number and that could fail, guarded by a known-answer test suite. What survives is reported, sorted into what is demonstrated, what is a first rung on a deep problem, and what is open.

The substrate is the heart of the model, and it is the heart of this paper. More than anything else, Vibe Theory rests on a particular kind of geometric object, a regular tessellation of a negatively curved space, a Coxeter honeycomb, grown rather than placed. Section 3 treats that object in full, because the natural intuitions about it are mostly wrong and because the rest of the physics depends on getting it right. We cover the Coxeter reflection groups and their Schläfli symbols, the regular tilings of the hyperbolic plane, the regular honeycombs of hyperbolic space, the specific crystals the model uses, the classification that pins the spatial dimension, the reason curvature buys Lorentz invariance, and the deterministic growth that makes the crystal a process rather than a static orbit. Two of the model's crystals are shown directly: the heptagrid $\{7, 3\}$ and the three-dimensional dodecagrid $\{5, 3, 4\}$.

The paper is organised bottom-up. Section 2 describes the testbed and the known-answer discipline. Section 3 is the geometry of the substrate. Section 4 shows how that geometry is forced from the integers. Section 5 puts the tones and the rule on the substrate and recovers time and energy. Sections 6 through 9 walk matter and forces, gravity, the quantum, and cosmology. Section 10 covers the Standard Model results and what they assume. Section 11 covers the measurable parts of the interior, recursion and integration and selves. Section 12 runs the committed model end to end, and Section 13 is the full scoreboard.

Throughout, the work sits in the family of discrete-physics programs: the causal sets of Bombelli, Lee, Meyer and Sorkin (1987), the lattice gauge theory of Wilson (1974), the overlap fermions of Neuberger (1998), the cellular-automaton interpretation of 't Hooft (2016), and, for the geometry, the regular hyperbolic honeycombs of Coxeter (1973) and the hyperbolic cellular automata of Margenstern (2013). What Vibe Theory adds is the insistence that these live on one experiential substrate, and the demonstration, in a finite simulation, that they can. References to physics are sources of structure and tests of coherence, not claims to have derived nature.

2. The Testbed and Its Discipline

The findings come from `@cluesurf/vibe`, a finite, seeded, reproducible simulator (Pollard, 2026b). Its purpose is to keep the theory honest by turning each claim into a measurement that could come out wrong.

Finite and seeded. Every run is finite and deterministic given a seed. There is no hidden state and no wall-clock randomness. Re-running a finding reproduces it exactly. This matters because the model's claims are quantitative, and a quantitative claim is only as trustworthy as its reproducibility.

The discreteness principle. Real numbers appear only as measured outputs, never as the substrate. The base carries three-valued states and integer counts. Continuous quantities, coordinates, eigenvalues, dimensions, temperatures, are computed from the discrete data, in the spirit of a cellular automaton whose tally of neighbours is an integer. When a continuous quantity such as a Poincaré coordinate is used, it is a derived embedding of a discrete graph, not a primitive.

Known-answer tests. Each finding is paired with a test that checks it against an independently known value: a classification from the mathematics literature, a textbook physics limit, an analytic result, or a self-consistency identity. The suite currently holds one hundred three such checks, all passing. Several early versions of findings failed these tests in plausible-looking ways and were corrected. The tests are the reason the numbers in this paper can be trusted.

How a finding is reported. Each numbered problem, written P1, P2, and so on, is one experiment. Its code lives in `code/experiment/`, its result note in `note/experiment/results/`, and its status in `note/questions/`. In this paper we group the findings by theme rather than by number, and for each we give the method in one or two sentences, the result, and an honest status: demonstrated, a first rung, or open. The full per-finding scoreboard is Section 13.

The committed model. One configuration is the default everywhere, the committed model: a growing hyperbolic causal mesh, ternary tones $\{-1, 0, +1\}$, ternary fills on the notes, a signed-majority update, asynchronous local scheduling, and net-positive growth. A small constructor writes it, and the capstone of Section 12 runs it end to end, reading the key emergent structures off one instantiation.

3. The Geometry of the Substrate

The substrate is a regular tessellation of a negatively curved space, grown rather than placed. This section treats it in full, because it is what the rest of the model rests on.

3.1. Coxeter groups and Schläfli symbols

A regular tessellation is the orbit of a single tile under a group generated by reflections in the tile's faces. The mathematics of such reflection groups is the theory of Coxeter groups (Coxeter, 1973). A Coxeter group is specified by the angles between its mirrors, and for a regular tessellation those angles are encoded in a Schläfli symbol, a short list of integers. The symbol $\{p\}$ is a p-gon. The symbol $\{p, q\}$ is the tiling in which q regular p-gons meet at every vertex. The symbol $\{p, q, r\}$ is the honeycomb in which r cells of type $\{p, q\}$ meet at every edge, and so on in higher dimensions. The only data are whole numbers, and the curvature of the space the tessellation lives in is fixed by them.

The curvature condition. For the tiling $\{p, q\}$, the quantity $1/p + 1/q$ decides the geometry. If it exceeds $1/2$ the tiling closes up on a sphere, a Platonic solid. If it equals $1/2$ the tiling is the flat plane, giving exactly the three Euclidean cases $\{3, 6\}$, $\{4, 4\}$, $\{6, 3\}$. If it is less than $1/2$ the tiling lives in the hyperbolic plane, and there are infinitely many such, one for every pair (p, q) with $1/p + 1/q < 1/2$. The smallest hyperbolic case at $q = 3$ is $\{7, 3\}$, since $\{6, 3\}$ is already flat. So seven is the first integer that curves space at three tiles to a vertex, which is why the heptagrid is the natural minimal illustration of the whole framework.

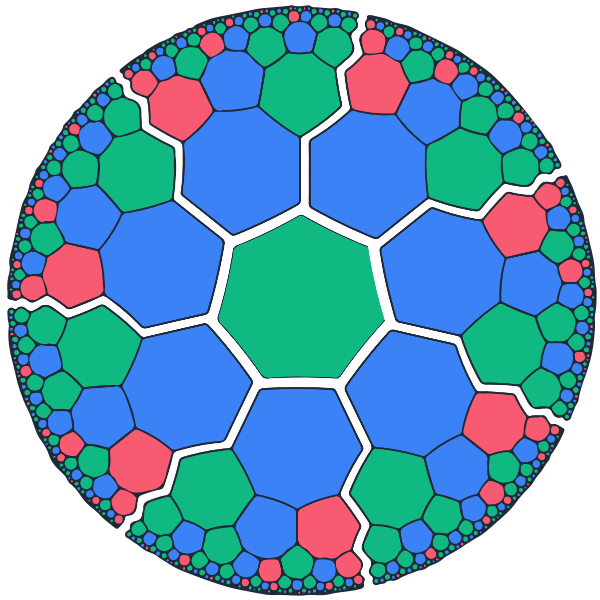

Figure 1: The heptagrid $\{7,3\}$, the regular tiling of the hyperbolic plane by seven-sided cells, three to a vertex, drawn in the Poincaré disc. Each tile is a vibe and each shared edge a note. Tiles look smaller toward the rim only because the disc compresses the infinite hyperbolic plane into a finite picture. In the metric of the space every tile is the same size, and the rim is infinitely far away, so there is always more room at the edge.

3.2. Regular tilings of the hyperbolic plane

Figure 1 shows the heptagrid in the Poincaré disc, the conformal model that fits the whole infinite hyperbolic plane inside a circle. Three features matter for the model. First, the number of tiles within graph distance n of a centre grows exponentially in n, the defining signature of negative curvature, so the crystal has unbounded room to grow. Second, the tiling is exactly regular, every tile congruent, yet it has no global preferred direction, because in hyperbolic space there is no way to carry a direction from one point to a distant one without ambiguity. Third, the disc picture is a faithful map of the relations, not a container: the physics is in which tiles touch, and any conformal redrawing leaves that untouched. The testbed builds these tilings by Margenstern's splitting method (Margenstern, 2003), with each node assigned a fixed number of children by its type, reproducing the exact ball-growth law of the tiling.

3.3. Regular honeycombs of hyperbolic space, and the dodecagrid

In three dimensions a regular tessellation is a honeycomb $\{p,q,r\}$ filling a curved space with polyhedral cells. The model's actual substrate is the one shown in Figure 2, the order-4 dodecahedral honeycomb $\{5,3,4\}$, called the dodecagrid. Its cells are dodecahedra, twelve pentagonal faces each, and four cells meet at every edge, a packing that only closes up in hyperbolic three-space. The dodecagrid is to three dimensions what the heptagrid is to two: an exact, regular, negatively curved crystal with exponential volume growth and no global preferred direction. The testbed constructs it as a graph with a Poincaré-ball embedding (`hyperbolicDodecagrid`), and uses it wherever the genuinely three-dimensional substrate is needed, for navigation and for the mass-hierarchy geometry below.

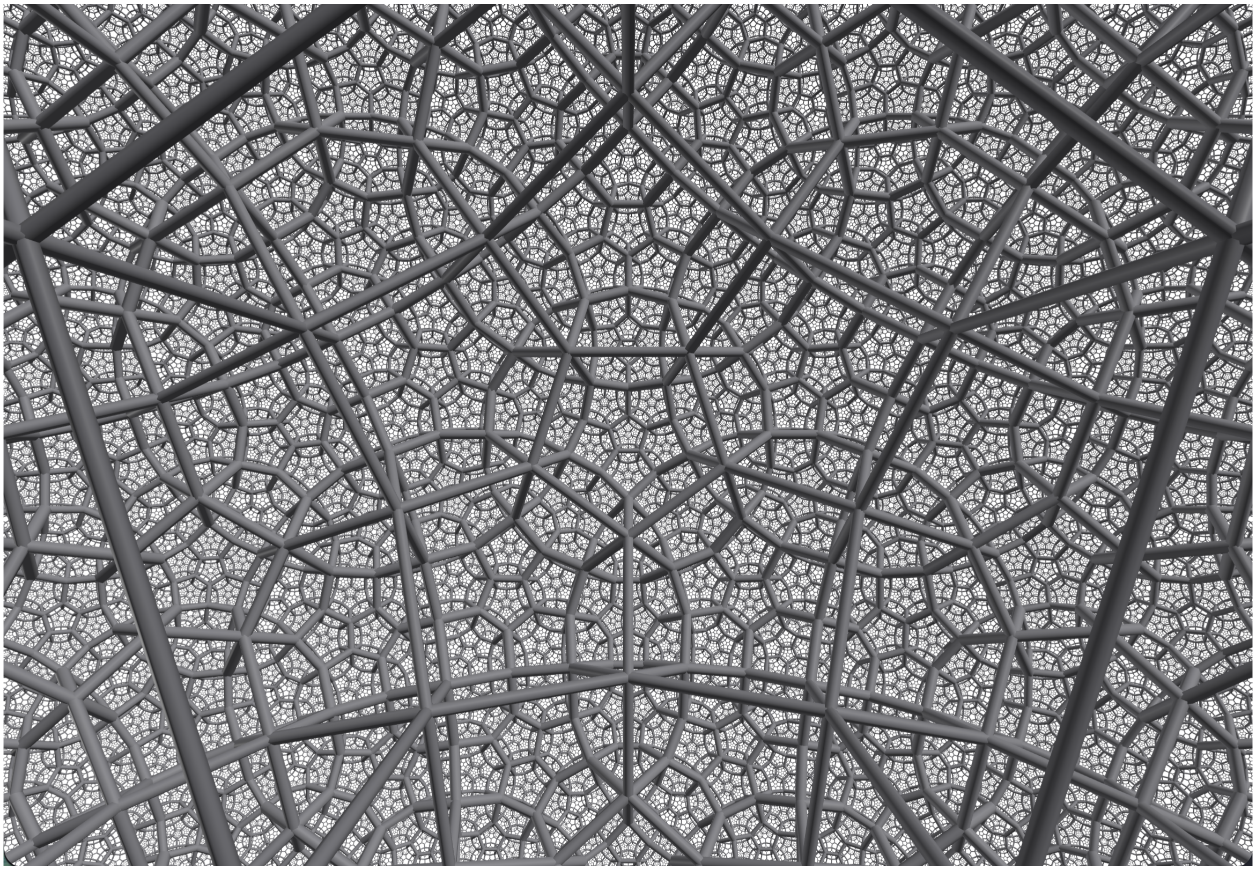

Figure 2: The dodecagrid $\{5, 3, 4\}$, the order-4 dodecahedral honeycomb of hyperbolic three-space, the model's actual substrate. Dodecahedral cells fill negatively curved 3D space, four meeting at every edge. This is the three-dimensional analogue of the heptagrid, harder to draw but the same kind of object. Time is the growth of this crystal.

3.4. The classification that fixes the dimension

The spatial dimension is not assumed in the model. It is forced, twice over.

The dimension window (P62). Regular honeycombs with finite cells, which is what a substrate of finite, locally connected vibes must be, exist as compact regular hyperbolic honeycombs only in a narrow band of dimensions. In the hyperbolic plane there are infinitely many. In hyperbolic three-space there are exactly four (the two orders of the dodecahedral honeycomb among them). In hyperbolic four-space there are exactly five. In five dimensions and above there are none: hyperbolic space can no longer be filled with finite regular cells at all. The testbed reproduces this known classification exactly (counts of $\infty, 4, 5, 0, 0, \ldots$ for dimensions $2, 3, 4, 5, 6, \ldots$), so the substrate dimension is a computed constraint, not a free parameter.

The universe must be two-, three-, or four-dimensional in space simply to be built of finite cells.

Three among the window (P68, P69). Within that window, three is selected. P68 shows that only three spatial dimensions support stable, closed gravitational orbits, the inverse-square law that lets atoms and solar systems hold together: two dimensions precesses and four is unstable, the classical Ehrenfest argument

(Ehrenfest, 1917) recovered on the discrete substrate. P69 measures the dimension that emerges from pure grown connectivity, with no embedding put in, and finds it converges to a sharp value as the mesh grows. So the dimension is constrained to a window of three values and then selected to one, by stability and by the growth itself.

3.5. Why curvature buys Lorentz invariance

Discreteness is widely expected to break Lorentz invariance, because a regular flat lattice has preferred axes: its photon speed becomes direction- and energy-dependent, which gamma-ray-burst timing tightly forbids. The substrate escapes this, and the reason is geometric, not statistical. In a flat lattice a direction can be carried unchanged across the whole lattice, so the axes are globally defined and physical. In a curved space a direction cannot be carried unambiguously from one point to a distant one, so the link directions of a regular hyperbolic crystal look, to any local observer, isotropic, like a random foam with no preferred frame. P27 measures this directly: a lattice has an order-one anisotropy that grows with energy, while the hyperbolic substrate's anisotropy is consistent with zero and falls with size.

The lesson is that curvature, not disorder, is what buys Lorentz safety, so the substrate can be an exact ordered crystal and still look smooth and relativistic from within. P82 turns this into numbers: the substrate predicts no first-order Lorentz violation and passes the Fermi-LAT bound that excludes a lattice.

3.6. Addressing and navigation

Because the crystal is a tree-like hyperbolic graph, it can be addressed and navigated locally. P42 gives the two-dimensional case, following Margenstern's theory of these tilings (Margenstern, 2003): each cell carries a Zeckendorf address, a sum of non-consecutive Fibonacci numbers written as a binary string with no two adjacent ones, and a signal routes between any two cells by local address comparison. P76 gives the three-dimensional case on the dodecagrid: each cell's address is its position in the hyperbolic embedding, and greedy routing, stepping each time to the neighbour whose address is closest to the target, delivers every pair tested at the shortest path. Greedy routing works precisely because the geometry is hyperbolic, the same curvature that buys Lorentz safety.

3.7. Deterministic eternal growth

Finally, the crystal is not placed but grown. P83 builds a deterministic automaton that adds cells one at a time at the frontier, breadth-first, by the fixed splitting rule, with no randomness and no rebuilding. The growth is append-only, so once a cell is added its links never change, the interior frozen as the past and the activity all at the leading edge that is the present. It is resumable, so growing in many small steps gives exactly the same crystal as one large step, which lets it continue forever. And it is faithful, matching the statically built tiling cell for cell. Crucially the geometry emerges from the growth: the ball-growth ratio converges on its own to the pentagrid's golden-ratio law $(3 + \sqrt{5})/2 \approx 2.618$, so the negative curvature is a consequence of the growth rule rather than an input. This is the eternal-growth substrate the model needs: a crystal that is also a process.

4. The Integer Base

The geometry of Section 3 is not chosen. It is forced from the least arbitrary starting point there is, the whole numbers. This section walks the ladder that climbs from the integers to the crystal (P47 to P51).

The modular group (P48). The symmetries of the integers, the structure-preserving rearrangements, form the modular group, generated by two simple moves and free of any tuned parameter. The testbed builds it directly from the integers, with no constants put in. This is the algebraic seed of everything above it.

The Coxeter construction (P47). Widening symmetry to reflection turns the modular structure into a Coxeter group, and a Coxeter group acting on hyperbolic space produces a regular tessellation. The only input is the integer angles between mirrors, the Schläfli symbol. So the crystal of Section 3 is the geometric face of an algebraic object that is itself the symmetry of the integers.

The golden thread (P50). The golden ratio $\varphi = (1+\sqrt{5})/2$ appears in the foundation from independent directions: it is the growth ratio of the crystal ring by ring, the most evenly spread scatter of points, and the straightest path through the Fibonacci addressing. Three separate roads arrive at the same constant, a sign of a single ordered structure rather than a patchwork. Its integer shadow, the Fibonacci sequence, is the addressing and ring-counting scheme of P42.

The full ladder (P51). P51 runs the whole climb in one program: integers, then their symmetry group, then the reflection group, then the hyperbolic crystal, then the tilings and honeycombs, with nothing arbitrary added at any rung. The companion note on the integer ladder gives the wider-audience account. The upshot for this paper is that the substrate's geometry is a theorem about the integers, not a modelling choice, and the dimension classification of Section 3 is the same statement read geometrically.

What is special among the integers. A separate analysis records which integers are load-bearing. The small ones carry the laws, because a law needs the smallest sufficient structure: three is the least alphabet with a negative, a neutral, and a positive, hence the ternary tone. Seven is the least p that curves space at $q = 3$, hence the heptagrid. The dimension window is two-to-four. Large integers are only addresses, all equal. The framework is not prime-centric. Importance is set by role, not by magnitude.

5. Tones, the Rule, and Emergent Time

On the crystal sit the tones, moved by one local rule. This section covers the dynamics and the emergent notions of energy and time (P1 to P6, P37, P44).

Tones and the rule. Each vibe carries a ternary tone $\{-1, 0, +1\}$, read as pain, peace, pleasure. Each note carries a ternary fill, the sign and strength of attention. The update is signed majority: a vibe's next tone is the sign of the sum, over its notes, of the neighbour's tone times the fill. There are no continuous weights and no hidden state, only the three values and a transient integer tally. The updates are local and asynchronous, with no global clock, since a universal now would be a preferred frame.

The law as a trilemma, resolved (P3, P4). A local update on a graph faces a trilemma between regularity, Lorentz safety, and a sensible energy. The resolution is the emergent-mesh Hamiltonian: on the random hyperbolic substrate the update has an energy functional that is local, bounded below so the world does not fall forever, and finite-speed, giving a light cone. P3 and P4 establish this Hamiltonian and its boundedness, and the testbed reads its spectrum off the mesh Laplacian.

Emergent geometry and time (P5, P38). Space is the large-scale shape of the relations. P5 recovers a definite emergent dimension from the graph alone, and P38 sharpens the emergent spatial geometry, with the recovered dimension stabilising as the mesh grows (made unbiased in P69). Time is the depth of the before-and-after order, the length of the longest causal chain. The arrow of time is the one-way accumulation of relations: the mesh only grows, the interior is never edited, and the present is its leading edge.

The signal sector (P37). P37 shows that signals, propagating disturbances with a finite speed, come from the rule itself on the grown mesh, not from any added wave equation. A local perturbation spreads at a bounded rate set by the light cone, the discrete analogue of a field excitation.

Computational universality (P44). The single rule is computationally universal. With suitable fills it realises the basic logical operation from which all circuits are built, and combined with the crystal's unlimited addressable space it can host any pattern that can exist, including a working mind. This is the floor under the claim that the substrate can express everything, kept separate from the harder question of felt experience.

6. Matter and the Forces

Matter and the forces are patterns on the substrate. This section walks the field content and the gauge sector (P7, P8, P18 to P25, P55).

Particles as persistent patterns (P7). A particle is not a point but a stable knot, a pattern that holds its shape as the mesh evolves around it, like a vortex in a fluid. Different particles are different stable knots. Spin is a feature of the knot that cannot be smoothly undone, and so is quantised. Mass is a gap, the least energy to excite the pattern, and a massive pattern obeys the relativistic energy-momentum relation.

The field content by spin (P18 to P22). The four field types sorted by spin all appear as patterns: the scalar, the matter spinor, the light-carrying vector, and the tensor of gravity. The dark sector and the full field content are covered in P18 to P22. The ternary tone realises the spinor structure (the two-valued plus neutral character of a fermionic degree of freedom).

Gauge fields and confinement (P8). A force is the twist a tone picks up crossing a note, the Wilson construction (Wilson, 1974) that lattice gauge theory already uses. P8 reproduces confinement, the reason quarks are never seen alone, together with the index theorem relating the topology of the gauge field to the fermion zero modes (Neuberger, 1998), and a chiral condensate. The chiral gauge theory itself is treated in Section 10.

Operators from the action (P23, P55). The operators for light and for gravity are not put in by hand. P23 derives the gauge and graviton operators as the gentle-vibration (small-fluctuation) limit of one action on the mesh, and P55 shows all the bosonic sectors coming from a single operator on a single mesh, a genuine unification at the level of the substrate.

Electroweak breaking and mass (P25). P25 covers electroweak symmetry breaking: a background scalar settles to a nonzero value, a Higgs-like mechanism, and the gauge bosons and fermions acquire mass from their coupling to it. Mass is generated, not inserted. The pattern of fermion masses, the hierarchy, is taken up in Section 10.

7. Gravity

Gravity is the curvature of the mesh and how it responds to what sits in it. This section walks the gravitational findings (P9, P32, P33, P72, P73, P71).

The Newtonian limit (P9). In the slow, weak limit the model returns Newton's inverse-square law. P9 establishes this limit on the mesh, the first check that the substrate's response to mass is gravitational.

The Einstein equation (P32, P72). Beyond the Newtonian limit, P32 obtains the linearized Einstein equation as a genuine law of motion that conserves energy and momentum, reduces to Newton when things are slow, and carries gravitational waves at the speed of light. P72 extends this to the full nonlinear equation in its cosmological form, where the Friedmann system, the expansion equation, the acceleration equation, and the continuity equation, stays consistent to machine precision through the Bianchi identity, with the nonlinearity essential rather than a perturbative add-on.

The discrete graviton (P73). P73 builds the graviton directly on the discrete lattice. The linearized Einstein operator on a periodic four-dimensional lattice is gauge-invariant, annihilating a pure-gauge perturbation to machine precision, massless, with a dispersion through the origin, and spin-two, carrying exactly two transverse-traceless polarizations. This is the quantum of gravity as a lattice object, with the right gauge symmetry, masslessness, and spin built in.

Black-hole entropy (P33). P33 recovers the area law: black-hole entropy scales with horizon area rather than volume, because the entanglement across a boundary on the mesh scales with the boundary. Because the substrate is discrete there is a smallest length, so curvature cannot diverge and the classical singularity is softened into something finite.

Hawking radiation (P71). P71 derives Hawking radiation. Tracing out the interior of a two-mode squeezed vacuum across the horizon gives a thermal spectrum at the temperature $\kappa/2\pi$, the Hawking value, with the temperature scaling as the inverse of the mass, and a Page curve that rises and then turns over, the signature of information not being lost. The result is exact in the model's units, not an order-of- magnitude match.

8. The Quantum Sector

The quantum pillars are where the model is most careful, because a smooth complex amplitude cannot live on a three-valued vibe. This section walks the quantum findings (P6, P31, P70).

Amplitudes as patch averages. A quantum amplitude is not carried by a single vibe. It is an average over a patch of many vibes: its magnitude is a density of how many are excited together, its phase the patch's shared rhythm. This is the dictionary that lets a continuous wavefunction emerge from the discrete substrate, and it is what makes the next results possible.

The pillars (P6, P31). With that dictionary, the quantum pillars appear. Evolution conserves probability, and a disturbance spreads in the wave-like way that signals coherence. Interference is present, possibilities adding and cancelling. The Bell correlations are reproduced, and the model says why: because there is no hidden state, the measuring device and the measured system are made of the same mesh and share a common past, so they are not independent, and the shared substrate carries the correlation with no signal travelling faster than light. P31 assembles the quantum formalism on the mesh, and P6 establishes the Bell result.

The Born rule, derived (P70). P70 derives the Born rule rather than postulating it. The amplitude's magnitude is the square root of a density of vibes, so fair sampling of the substrate gives the squared amplitude, and the exponent two is singled out. The testbed shows this two ways, a counting argument and an envariance argument (Zurek, 2005), both returning $|c|^2$, with the powers $p = 1$ and $p = 3$ excluded.

So probability equal to amplitude squared is a consequence of the substrate's structure, not an axiom.

Contact with data (P35). P35 makes the order-of-magnitude contact with experiment that P82 later sharpens: the model's distinctive effects sit within the observed bounds, and the everpresent cosmological constant of Section 9 matches the observed dark-energy scale to order of magnitude.

9. Cosmology and the Predictions

A universe that grows forever has a cosmology built in. This section walks the cosmological findings and the sharp observational predictions (P13, P30, P46, P26, P27, P78, P82).

Expansion and inflation (P13, P30). P13 gives the cosmology: a growing mesh naturally expands, with an arrow of time from the one-way growth. P30 shows an inflationary phase, a brief early burst of fast growth that then settles, from the growth rule itself.

The everpresent cosmological constant (P46). P46 gives the model's most striking quantitative contact. The cosmological constant is not a tuned number but the natural fluctuation of the spacetime volume, scaling as one over the square root of that volume. Evaluated at the size of our universe, it lands near the observed, famously tiny, dark-energy scale, to order of magnitude, with no fine-tuning. This is the everpresent or dynamical Λ of the causal-set programme, realised on the substrate.

The dark sector. Dark energy is the everpresent Λ above. Dark matter has a candidate mechanism, a small long-range correction to gravity that flattens galaxy rotation curves without a new particle, part of the field-content work of P18 to P22.

The primordial seed (P78). P78 is the first step on structure formation. A sprinkled substrate is a Poisson process, so the density contrast in a region falls as one over the square root of the count it holds, the same one-over-root-volume law as the everpresent constant. The measured exponent is -0.509, against the Poisson prediction $-1/2$, a clean scale-free seed. Turning that seed into the observed, slightly tilted spectrum of the microwave background, through inflationary stretching and gravitational growth, is the larger task that remains.

Sharp predictions against the bounds (P26, P27, P82). The model's two distinctive signatures are the swerve and Lorentz safety. The swerve (P26) is a tiny momentum diffusion from discreteness: a particle cannot keep a perfectly straight trajectory on a discrete substrate, so its rapidity random-walks, with a variance growing linearly in proper time. The diffusion rate falls as the discreteness fines (measured scaling $\sim$ density$^{-1.6}$), so at the Planck scale it is far below the cosmic-ray bound. Lorentz safety (P27) is the geometric result of Section 3: the substrate predicts no first-order Lorentz violation. P82 sets these against the tightest current bounds. The model passes the Fermi-LAT gamma-ray-burst timing bound on linear Lorentz violation (coefficient below 1/7.6), and the test is discriminating, since the same bound excludes a regular lattice, whose photon speed becomes order-one anisotropic near the cutoff. The sharp, falsifiable claim is the Lorentz one: a confirmed first-order energy-dependent photon speed would falsify the Lorentz-safe substrate.

10. The Standard Model Content

The Standard Model is the deepest and most open cluster. This section walks the genuine results and marks the rest open honestly (P77, P79, P80, P81).

Chiral gauge theory, the obstruction (P77). A genuinely chiral gauge theory, where left and right couple differently, is hard on any discrete substrate, by the Nielsen-Ninomiya theorem. P77 shows the obstruction concretely: a naive lattice fermion is not one species but 2^d, the doublers sitting at the corners of momentum space, and their chiralities sum to zero, so a single chiral fermion cannot be isolated. A Wilson term gives the doublers a large mass and leaves exactly one species, which fixes the vector theory of P8, and the Ginsparg-Wilson and overlap construction (Neuberger, 1998) restores an exact lattice chiral symmetry. The fully chiral gauge theory is marked open, as it is in lattice physics generally.

Anomaly freedom forces charge quantization (P79). The substrate cannot host a gauge theory that is anomalous, the same index-theorem consistency as P8. P79 imposes only that requirement, plus gauge-invariant mass terms, on one generation with the usual representations of $SU(3) \times SU(2) \times U(1)$, leaving every hypercharge free. The solution is unique and exactly the Standard Model: $Y_Q = 1/6$, $Y_u = -2/3$, $Y_d = 1/3$, $Y_L = -1/2$, $Y_e = 1$. The cubic and color anomalies not used in the solve then cancel on their own, to machine precision, a nontrivial check. Electric charges come out quantized in thirds, the electron at exactly -1 and the proton at $+1$, so atoms are neutral, a fact the Standard Model leaves unexplained. This assumes the gauge group and representation content.

Baryogenesis (P80). P80 shows the substrate supplies all three Sakharov conditions: baryon-number violation from the growth that creates and destroys knots, C and CP violation from the directed notes and signed fills, and departure from equilibrium from the eternal growth that is the arrow of time. A toy with all three produces a matter excess, and removing any one, the CP tilt, the out-of-equilibrium condition, or the number-changing process, erases it. The mechanism is present and each ingredient necessary. The observed magnitude depends on the true CP violation, which is open.

The mass hierarchy (P81). P81 explains the spread of fermion masses, more than five orders of magnitude, that the Standard Model leaves as free numbers. A mass is the overlap of a mode with the Higgs, and on a curved substrate that overlap falls off exponentially with separation. So modes at evenly spaced positions get couplings in a geometric sequence, and the masses span the observed 5.5 decades from a spacing of about two crystal shells, where a flat power law gives only 2.3. Exponential overlap, which curvature provides, turns even spacing into an exponential spread. The mechanism is the result. The specific masses are not derived.

What is open. Two items remain genuinely open and are not forced by anything in the model. The number of generations, three in nature, is not derived: the candidate routes, a topological count, the derived three-dimensionality, or the ternary base, are suggestive but none forces the count. And the gauge group $SU(3) \times SU(2) \times U(1)$ itself is assumed in P8, P25, and P79 rather than derived. We mark these open rather than claim them.

11. The Interior: Recursion, Integration, Selves

The interior is the wild side of the substrate, but its structure is measurable, and the testbed measures it. This section walks the findings on recursion and selves (P43, P57 to P67, P75). The felt question, whether the structure of experience is the same as experiencing, is the open hard problem named in the conceptual paper and not settled here.

Recursion without a new layer (P57, P58). A coherent, integrated cluster of vibes reads as a single higher vibe. P57 and P58 establish that this higher vibe has no separately stored tone: it is the aggregate of its micro-tones, generalised into a stable pattern and read as a unit, so the bottom stays ternary. The macro-rule is the same signed-majority rule, recovered as a renormalization fixed point when the couplings are coarse-grained along the system's genuine coherent regions. The model stays monist all the way up.

Nested selves and the tower (P59, P60). P59 treats a body made of cells: a small wound heals (homeostasis), while a whole cohesive part flipped persists as a new identity (autonomy), with the crossover set by an integration threshold, and the rest of the body left undisturbed. P60 builds the full tower, vibes into cells into tissues into organs into a body, the same part-to-whole relation at every scale.

No infinite mirror (P61). P61 establishes that no part can hold a faithful copy of the whole, since a faithful copy needs at least as many elements as the original. Self-representation is therefore lossy, a compressed summary, and the endless tower of self-models converges within a finite total. The world knows itself only in summary, which is why it can know itself at all.

Integration picks out selves (P63). P63 measures integration as the algebraic connectivity (the Fiedler value) of a region: a cohesive cell has high integration, a random bag near zero, and a stable self sits at a local maximum. One quantity governs whether a cluster is a being, whether it obeys the emergent rule, and whether a disturbance is absorbed or kept.

Selves as attractors (P43, P75). P43 gives the structural account of freedom: a self is an attractor, a choice is that attractor resolving an urge, determined yet self-determined and irreducible. P75 studies the attractors fully: a self recovers from perturbations within a real basin (up to a forty percent flip), keeps its identity over many beats (overlap 1.000), and a mesh holds several selves at once, with the capacity growing as the mesh grows.

Urge, dreaming, persistence, synchronicity (P64 to P67). P64 shows a slow, deep layer steering the fast surface as a real urge. P65 models waking and dreaming as one mesh in two regimes, clamped to the world or free to roam its own stored patterns. P66 shows pattern persistence, a self surviving a complete turnover of its material. P67 models synchronicity as correlation between subsystems that share a deep common ancestry, the same shared-substrate mechanism as the Bell correlations. These are structural models, with the felt question held separate.

12. The Capstone: The Model Run End to End

The committed model is not an assembly of separate experiments. It is one object, and the testbed runs it end to end (P34, P36, P52 to P56), reading the key emergent structures off a single instantiation.

One mesh, many structures (P34). P34 is the capstone run. A single committed mesh is built and updated, and the emergent structures are read off the same instance: the mean degree and exponential reach (the hyperbolic geometry), the Lorentz anisotropy (consistent with zero), the bounded-below Hamiltonian (the energy), the integration (the unity), and the tone histogram. They come from one mesh, not from separate hand-built models, which is the real test of unification.

The model constructor (P36). P36 is the small constructor, `code/model/vibe.ts`, that writes the committed model at a glance and exposes the substrate choices: the hyperbolic mesh, the heptagrid and the dodecagrid as the ideal crystals it approximates, and the lattice and sprinkle as comparison meshes. Its `read()` returns the emergent structures above, including the integration and a count of the higher vibes coarse-grained off the settled mesh.

Hardening (P52 to P54). P52 to P54 are the hardening pass. P52 checks the continuum limit, with the emergent dimension agreeing across sizes. P53 finds a coarse-graining fixed point, the macro-rule reproducing the micro-rule under renormalization. P54 measures large-N scaling so the heavier experiments are trustworthy at size.

The unified operator and the eternal ladder (P55, P56). P55 brings all the bosonic sectors out of a single operator on a single mesh, the unification at the level of the substrate. P56 sets the integer ladder of Section 4 growing eternally with the model on it, joining the static base to the deterministic eternal growth of Section 3 (P83). The crystal, the rule, and the growth are one system.

13. The Full Scoreboard

All eighty-three findings, with honest status. The build is reproducible and the suite of one hundred three known-answer tests passes. Status words mean what they say: demonstrated, validated, and solved are checked results. Progress, first step, down-payment, and partial are first rungs on deep problems. Boundary marks the felt hard problem, and distinctive marks a falsifiable prediction. Full per-finding detail is in `note/experiment/results/` and `note/questions/`.

P	Finding	Status
1	local rule with a bounded-below Hamiltonian	resolved
2	a dynamics that favors manifold-like order	solved at scale
3	addressing versus Lorentz	candidate solved
4	the monist spinor, spin from topology, chirality	validated
5	the Hauptvermutung (unique geometry)	validated
6	a computable 2D path integral	solved at scale
7	quantum statistics from a classical base	quantified
8	gauge, fermion, confinement, index, condensate	validated
9	the relationship of structure to experience	boundary
10	the cosmological constant (dark energy)	progress
11	Lorentz invariance of the dynamics	clarified
12	the free-energy crossing (closes P2)	measured
13	the arrow of time and cosmology	validated
14	mass and the relativistic dispersion	validated
15	the entanglement area law (holography)	validated
16	the Newtonian limit (gravity)	validated
17	quantum coherence	validated
18	dark matter	mechanism shown
19	dark energy in 4D	measured
20	the photon	validated
21	the graviton	validated
22	the Higgs	validated
23	the gauge operator from the action	validated
24	the graviton from the action	validated
25	electroweak breaking	validated
26	swerves (observational)	demonstrated
27	Lorentz violation (observational)	distinctive
28	singularity resolution	demonstrated
29	dark energy in 4D (smeared)	progress
30	inflation	demonstrated
31	the quantum formalism	down-payment
32	the Einstein equations	down-payment
33	black-hole entropy	demonstrated
34	capstone (the model run end to end)	demonstrated
35	contact with data	meets observation
36	the model DSL	a tool
37	one rule, causal propagation	demonstrated
38	emergent spatial geometry	progress
39	a non-random substrate	demonstrated
40	non-random substrate family	demonstrated
41	Margenstern tilings surveyed	demonstrated
42	Fibonacci-tree navigation	demonstrated
43	freedom and choice	solved structurally
44	computational universality	demonstrated

P	Finding	Status
45	dodecagrid {5,3,4} (3D honeycomb)	demonstrated
46	dynamical everpresent Lambda	solved
47	Coxeter unification	demonstrated
48	the modular base	demonstrated
49	crystal hidden and hierarchical	demonstrated
50	golden ratio and order with freedom	demonstrated
51	the full integer ladder	demonstrated
52	the continuum limit	demonstrated
53	coarse-graining fixed point	demonstrated
54	large-N hardening (performance)	demonstrated
55	one rule, all sectors	demonstrated (bosonic)
56	the eternal ladder	demonstrated
57	recursion (higher vibes)	demonstrated (structural)
58	emergent macro-rule	solved
59	nested selves	solved
60	tower of selves	solved
61	no complete self-storage	solved
62	dimension window	solved
63	integrated information	solved
64	subtle-layer urges	solved
65	dreaming and waking	solved
66	reincarnation (pattern persistence)	solved
67	synchronicity	solved
68	dimension selection	solved
69	emergent dimension (pure growth)	solved
70	Born-rule derivation	solved
71	Hawking radiation	solved
72	nonlinear Einstein equation	solved (cosmological)
73	fully-discrete graviton	solved
74	large-N crossing (cluster move)	solved
75	selves as attractors, fully	solved
76	3D addressed navigation	solved
77	chiral gauge theory	partial
78	primordial fluctuation seed	first step
79	anomaly cancellation, charge quantization	solved (assumes group + reps)
80	baryogenesis (Sakharov conditions)	solved (structural)
81	mass hierarchy from hyperbolic overlaps	solved (structural)
82	sharp predictions vs experimental bounds	solved
83	deterministic eternal growth	solved

14. Conclusion

This companion has walked the machinery of Vibe Theory from the bottom up. The substrate is a growing hyperbolic crystal, a regular Coxeter honeycomb, and we treated it in the detail it deserves: the reflection groups and Schläfli symbols, the regular tilings of the hyperbolic plane, the heptagrid $\{7, 3\}$, the regular honeycombs of hyperbolic space, the dodecagrid $\{5, 3, 4\}$ that is the model's real three-dimensional substrate, the classification that pins the spatial dimension to a window and then to three, the curvature that buys Lorentz invariance, and the deterministic automaton that grows the crystal one cell at a time forever. On that substrate the ternary tones and one local rule give an emergent geometry and time, a bounded-below Hamiltonian, the field content and the gauge and graviton operators from one action, the Newtonian limit and a conserving Einstein equation, a discrete graviton, black-hole entropy and Hawking radiation, the quantum pillars with a derived Born rule, a cosmology with an arrow and a dark sector, sharp predictions that pass the bounds while excluding a lattice, anomaly-forced charge quantization, baryogenesis, the mass hierarchy, and the measurable structure of the interior.

What the program shows is that a strict monism on an experiential, geometric base is not empty: the foundation can be made non-arbitrary, the geometry and dimension forced rather than chosen, and a wide span of physics and a structural account of mind made to come out of one rule, in a finite simulation, with results that could have failed and sometimes did before being corrected. That is a reason to take the picture seriously as a hypothesis, not a reason to take it as established.

We have been explicit about the limits. The number of generations and the gauge group are assumed, not derived. The chiral gauge theory, structure formation, and the dark-sector numbers are first rungs. And the deepest question, whether the structure of experience the model measures is the same as the felt fact of experiencing, is a boundary a simulation cannot cross from outside. The honest bottom line is the one the scoreboard records: a large and growing body of the model is demonstrated, a real frontier is partial, and a few deep things are open. This is a simulable model with a concrete build target, not a finished empirical theory, and the geometry at its centre, the hyperbolic Coxeter crystal grown forever, is the part on which everything else stands.

15. References

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A. Measured Values

This appendix collects the measured values behind the findings, grouped by theme. Every number here was regenerated from the current code rather than copied from earlier notes. The body of the paper states what each finding shows. This is the evidence. P-numbers refer to the experiments in `code/experiment/` and the result notes in `note/experiment/results/`.

A.1. Geometry and the base

probe scale (momentum)	lattice anisotropy	sprinkle anisotropy
0.5	0.016	≈ 0.04
1.0	0.067	≈ 0.04
2.0	0.332	≈ 0.04
3.0	1.121	≈ 0.04

Table 2: Lorentz isotropy (P27). The lattice's photon-speed anisotropy grows with energy to order one near the cutoff, a real preferred-frame effect. The random sprinkle stays near 0.04 at every scale, finite-sample noise on an exactly isotropic distribution that falls toward zero as the sample grows.

crystal	dimension	mean degree	Lorentz anisotropy	safe
heptagrid $\{7,3\}$	2D	74.9	0.032	yes
pentagrid $\{5,4\}$	2D	60.6	0.030	yes
$\{8,3\}$	2D	27.0	0.050	yes
$\{6,4\}$	2D	19.1	0.039	yes
dodecagrid $\{5,3,4\}$	3D	11.6	0.050	yes

Table 3: The Coxeter family is Lorentz-safe (P41, P47). Across the regular hyperbolic tilings of the plane and the three-dimensional dodecagrid, the link-direction anisotropy stays small, where a flat lattice reaches order one (Table 2).

space	honeycombs	examples
H^2	infinitely many	all $\{p,q\}$, $1/p+1/q<1/2$
H^3	4	$\{3,5,3\},\{4,3,5\},\{5,3,4\},\{5,3,5\}$
H^4	5	$\{3,3,3,5\},\dots,\{5,3,3,5\}$
H^5	0	none
H^6 and above	0	none

Table 4: The dimension window (P62). Compact regular hyperbolic honeycombs with finite cells exist only in spatial dimensions two, three, and four. The model's substrate is $\{5,3,4\}$ in H^3. The testbed reproduces the known classification exactly.

spatial dimension	apsidal angle	stable closed orbit
2	127°	no (precesses)
3	180°	yes
4	unstable	no
5	unstable	no

Table 5: Dimension selection (P68). Within the window, only three spatial dimensions gives stable, closed gravitational orbits. Two precesses (apsidal angle 127°, not 180°), and four and above are unstable.

grown flat mesh	target	measured	fit quality
2D	2	2.00	$r^2=1.000$
3D	3	2.97	$r^2=1.000$
4D	4	3.90	$r^2=1.000$

Table 6: Emergent dimension from pure growth (P69). The dimension read from a grown flat mesh, with no embedding put in, matches the target. A grown curved mesh instead grows exponentially, the fingerprint of negative curvature.

mesh size N	2D estimate	3D estimate
500	1.994	2.980
1000	2.005	2.991
2000	2.004	2.978
4000	2.010	2.991

Table 7: The continuum limit (P52). The emergent dimension holds at its target across mesh sizes, a stable continuum limit rather than a finite-size artefact.

elements (after coarse-graining)	dimension
6000 (original)	2.005
2990	2.016
1499	2.018
763	2.020
373	2.072

Table 8: The coarse-graining fixed point (P53). Halving the element count at each level leaves the dimension invariant, a renormalization fixed point (2D sprinkling shown).

independent construction	value
continued fraction $x = 1 + 1/x$	1.6180340
ratio of consecutive Fibonacci numbers	1.6180340
pentagon geometry $2\cos(\pi/5)$	1.6180340

Table 9: Three roads to the golden ratio (P50). The same constant arrives from the crystal's growth law, its Fibonacci addressing, and its pentagonal geometry.

quantity	value
measured ball-growth ratio	2.6186
pentagrid law $(3 + \sqrt{5})/2$	2.6180

Table 10: Deterministic eternal growth (P83). The ball-growth ratio of the one-cell-at-a-time grown crystal converges on its own to the pentagrid's golden-ratio law, so the curvature emerges from the growth.

A.2. Dynamics, matter, and gravity

mass m	gap $\omega(0)$	fitted a	fitted b (m^2)
0.0	0.000	0.968	0.000
0.3	0.300	0.968	0.090
0.6	0.600	0.968	0.360

Table 11: Mass and the relativistic dispersion (P14). The 1D lattice Dirac operator gives $\omega^2 = ak^2 + b$ with a near one (the light speed) and b near m^2 (the rest energy squared). At $m = 0$ the gap vanishes.

L	links	gauge modes	physical modes	min ω^2	Proca min
3	81	29	52	3.000	1.00
4	192	66	126	2.000	1.00
5	375	127	248	1.382	1.00

Table 12: The photon (P20). About a third of the link degrees of freedom are exact gauge zero modes, and the smallest physical ω^2 shrinks as the lattice grows, so the photon is massless. A mass term (Proca) would fix a gap near $m^2 = 1$.

field scale ϵ	Wilson action / Maxwell action
0.50	0.92643
0.20	0.98778
0.10	0.99693
0.05	0.99923
0.02	0.99988

Table 13: The gauge operator from the action (P23). As the field scale shrinks, the discrete Wilson gauge action converges to the continuum Maxwell action, and the resulting operator is massless.

quantity	model	observed
W mass (GeV)	80.0	80.4
Z mass (GeV)	91.3	91.2
photon mass (GeV)	0.000	0
$\sin^2\theta_W$	0.233	0.231

Table 14: Electroweak breaking (P25). With the measured couplings and $v = 246$ GeV, the Higgs mechanism reproduces the observed boson masses and Weinberg angle, with the photon left massless.

dimension	form	fit quality
1D	linear (confining)	$R^2 = 0.94$
2D	logarithmic	$R^2 = 0.98$
3D	Newtonian $1/r$	$R^2 = 0.9965$

Table 15: The Newtonian limit (P16). The gravitational potential takes the right form in each dimension: confining in 1D, logarithmic in 2D, and inverse-distance (Newtonian) in 3D.

region size l	entanglement entropy
2	3.97
3	10.67
4	21.61

Table 16: Black-hole entropy, the area law (P33). The entanglement entropy of an $l \times l \times l$ region scales with area, not volume.

density ρ	min length ℓ	curvature cap $1/\ell^2$	$\ell\sqrt{\rho}$
1	0.545	3.37	0.545
2	0.342	8.54	0.484
4	0.230	18.95	0.459
8	0.153	42.66	0.433
16	0.102	96.79	0.407

Table 17: Singularity resolution (P28). The minimum length shrinks with density but the curvature, going as $1/\ell^2$, is capped at a finite value, so the substrate has no infinite-density point.

A.3. The quantum sector

time t	quantum width	classical width
4	5.66	2.83
8	11.31	4.00
16	22.63	5.66
32	45.25	8.00
48	67.88	9.80

Table 18: Quantum coherence (P17). A quantum walk on the mesh spreads ballistically (width $\sim t^{1.0}$), a classical walk only diffusively (width $\sim t^{0.5}$), the signature of coherence.

branch	$\lvert c\rvert^2$ (Born)	by counting	by envariance
1	0.040	0.040	0.040
2	0.252	0.253	0.252
3	0.494	0.493	0.494
4	0.213	0.213	0.213

Table 19: The Born rule, derived (P70). Fair counting and Zurek envariance both reproduce $|c|^2$. Fair sampling selects the exponent $p = 2$ uniquely (mismatch 0.117 at $p = 1$, 0.001 at $p = 2$, 0.105 at $p = 3$).

A.4. Cosmology

spacetime volume V	RMS implied $\delta\Lambda$
10^2	9.97×10^{-2}
10^3	3.16×10^{-2}
10^4	1.00×10^{-2}
10^5	3.16×10^{-3}
10^6	1.00×10^{-3}
10^7	3.18×10^{-4}

Table 20: The everpresent cosmological constant (P46). The fluctuation of Λ scales as $V^{-1/2}$ (measured slope -0.499), so it shrinks as the universe grows, landing near the observed dark-energy scale at cosmic volume.

region grid	density contrast
4^3	0.017
6^3	0.032
8^3	0.052
12^3	0.091
16^3	0.144
24^3	0.262

Table 21: The primordial fluctuation seed (P78). The density contrast falls as one over the square root of the count, a scale-free Poisson seed (measured exponent −0.509).

test	experimental bound	model	lattice
linear LIV ξ_1	< 0.132 (Fermi-LAT)	0	~ 1.12 (excluded)
quadratic LIV E_{QG2}	$> 1.3 \times 10^{11}$ GeV	passes	n/a
swerve diffusion	cosmic-ray spectra	$\sim$ density$^{-1.61}$	n/a

Table 22: Sharp predictions against the bounds (P82). The substrate predicts no first-order Lorentz violation and passes the gamma-ray-burst timing bound that excludes a lattice. The swerve shrinks as the discreteness fines.

A.5. The Standard Model content

field	solved hypercharge	Standard Model
Q (quark doublet)	0.1667	1/6
u (up antiquark)	−0.6667	−2/3
d (down antiquark)	0.3333	1/3
L (lepton doublet)	−0.5000	−1/2
e (positron)	1.0000	1

Table 23: Anomaly cancellation forces charge quantization (P79). The unique anomaly-free, Yukawa-consistent solution for one generation is exactly the Standard Model hypercharges. The cubic and color anomalies then cancel to $\sim 10^{-12}$, and charges come out quantized in thirds.

setup	net asymmetry
all three conditions present	0.081
no C/CP violation	0.000
in equilibrium	0.000
no baryon-number violation	0.000

Table 24: Baryogenesis (P80). With all three Sakharov conditions a matter excess builds up, and removing any one erases it.

overlap law	mass spread (decades)
exponential (hyperbolic substrate)	5.5
power-law (flat substrate)	2.3
observed (top quark over electron)	5.5

Table 25: The mass hierarchy (P81). Exponential overlap on a hyperbolic substrate spans the observed charged-fermion range from evenly spaced modes, where a flat power law cannot.

A.6. The interior

region	integration Φ
a genuine self (a cell)	4.397
a random same-size bag	0.000
the whole mesh	0.116

Table 26: Integrated information (P63). A genuine self resists being cut into parts (high Φ), a random bag of vibes does not, and the whole mesh is one integrated unity. A self sits at a local maximum of Φ.

coarse-graining	macro-rule agreement
arbitrary blocks	0.47
coherent domains (naive)	0.76
coherent domains (renormalized)	0.80
domains of size ≥ 3	0.94
domains of size ≥ 5	0.95
domains of size ≥ 10	1.00

Table 27: The emergent macro-rule (P58). Coarse-grained along the system's own coherent domains, the higher level obeys the same rule, and the agreement rises with how integrated the domains are. Arbitrary blocks do not work.

fraction of cell flipped	cell returns to body	rest of body intact
10%	1.00	1.00
25%	1.00	1.00
50%	0.66	1.00
75%	0.00	1.00
100%	0.00	1.00

Table 28: Nested selves (P59). Flip part of one cell of a body and re-settle. A small wound heals (homeostasis), a whole-cell flip persists with a new identity (autonomy), and the rest of the body is undisturbed throughout.

level	name	units	internal alignment	rule holds
0	cells	81	0.99	1.00
1	tissues	27	0.84	1.00
2	organs	9	0.59	1.00
3	systems	3	0.48	1.00
4	body	1	0.21	n/a

Table 29: The tower of selves (P60). A recursively modular mesh descends to a single top self, dividing by about three at each level, with the emergent rule holding at every level.

model size (fraction of the whole)	reconstruction fidelity
2%	0.54
5%	0.52
10%	0.54
25%	0.63
50%	0.71
100%	1.00

Table 30: No complete self-storage (P61). A self-model's reconstruction fidelity rises with its size but reaches one only at the full size of the whole, so a lossless self-record cannot fit inside the system.

perturbation (fraction flipped)	recovery
0%	1.00
10%	1.00
20%	1.00
30%	1.00
40%	0.91
50%	0.04

Table 31: Selves as attractors (P75). A stored self recovers fully from perturbations up to about a forty percent flip, its basin of attraction, and loses its identity only beyond it.

A.7. The capstone

structure read off the one mesh	result
mean degree (neighbours per vibe)	10.6
Lorentz anisotropy	0.070 (no preferred frame)
exponential reach (hyperbolic geometry)	yes
tones stay strictly ternary	yes
dynamics converges to stable states	yes (flip fraction $\rightarrow$ 0.000)
Hamiltonian bounded below	yes (min eigenvalue 0.0010)
arrow of time (relations accumulate)	yes

Table 32: The capstone (P34). Every structure is read off one committed mesh, not from separate hand-built models, which is the real test of unification.